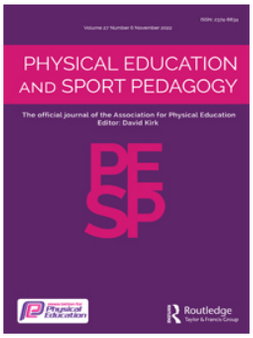

Physical Education and Sport Pedagogy

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/cpes20

Effects of gamified didactic with exergames on the psychological variables associated with promoting physical exercise: results of a natural experiment run in primary schools

Alejandro Quintas & Juan-Carlos Bustamante

To cite this article: Alejandro Quintas & Juan-Carlos Bustamante (2021): Effects of gamified didactic with exergames on the psychological variables associated with promoting physical exercise: results of a natural experiment run in primary schools, Physical Education and Sport Pedagogy, DOI: 10.1080/17408989.2021.1991905

To link to this article: https://doi.org/10.1080/17408989.2021.1991905

Published online: 22 Oct 2021.

Submit your article to this journal

Article views: 244

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=cpes20

PHYSICAL EDUCATION AND SPORT PEDAGOGY
https://doi.org/10.1080/17408989.2021.1991905

Routledge
Taylor & Francis Group

Check for updates

Effects of gamified didactic with exergames on the psychological variables associated with promoting physical exercise: results of a natural experiment run in primary schools

Alejandro Quintas [a] and Juan-Carlos Bustamante [b]

[a]Facultad de Ciencias Humanas y de la Educación, Universidad de Zaragoza, Huesca, Spain; [b]Facultad de Educación, Universidad de Zaragoza, Zaragoza, Spain

ABSTRACT

Background: Exergames are a new socio-technological phenomenon consisting of digital motor games that aim to stimulate players' motor skills. The physical-motor component of exergames has been well studied, but knowing more about their possible psychological effects in the school context is interesting to promote physical exercise practice. Gamification refers to the use of game-based elements in non-game contexts, whose aim is to motivate action by making activities more game-like. Gamification seems an innovative strategy to promote physical exercise habits at schools whose real effects are still poorly unknown. Physical education has significant direct and indirect influences on students' physical exercise. Exergames and gamification are associated with psychological benefits related to physically active behaviour.

Purpose: The aim of this study was to analyse the effects of a gamified exergaming school intervention on the psychological variables associated with physical exercise (PEx) promotion. We hypothesised that the gamified exergaming school intervention will produce more achievement motivation, more enjoyment, a more positive attitude towards exergames, and more exergame and PEx intention in students.

Participants and setting: A natural experiment with a non-randomised controlled design. The participants were 417 Primary Education students (53.2% girls, $n = 222$; 46.8% boys, $n = 195$) recruited from two public and two non-public schools (control group, $n = 191$; experimental group, $n = 226$). Their mean age was 11.1 (SD = 1.7), and 50.4% of the sample studied Primary Education Year 6 (aged 10–11 years; $n = 210$) and 49.6% studied Primary Education Year 7 (11–12 years; $n = 207$). The control treatment (traditional didactic intervention) was designed based on usual didactic dance teaching in Spain. Another similar experimental treatment was designed as the control treatment, but with a gamified climate and an exergame (gamified exergaming intervention). The Just Dance Now exergame was used. The Mechanics-Dynamics-Aesthetics framework was applied to achieve mastery-oriented motivational climates and an inclusive gamified atmosphere for all student-player types. Moreover, to gamify each class the ClassDojo application was used.

Data collection: The students completed: the Achievement Motivation in Physical Education Test (AMPET); the Sport Satisfaction Instrument (SSI); the 'Attitudes Toward Exergames' scale; the 'Exergame intention' scale; the 'Exercise Intention' scale.

ARTICLE HISTORY
Received 7 August 2020
Accepted 7 October 2021

KEYWORDS
Gamification; exergame; physical exercise; promotion; primary school

CONTACT Juan-Carlos Bustamante jbustama@unizar.es Facultad de Educación, Universidad de Zaragoza, calle Pedro Cerbuna, 12, Zaragoza 50009, Spain

Supplemental data for this article can be accessed https://doi.org/10.1080/17408989.2021.1991905.

© 2021 Association for Physical Education

Statistical analyses: In order to test the research hypotheses, both study groups were compared using factorial ANOVAs 2 (time; pre-treatment condition vs. post-treatment condition) × 2 (treatment; traditional didactic intervention vs. gamified exergaming intervention).
Findings: As the analysis threshold was set at ≤0.0125 with Bonferroni adjustment, the results showed better positive gamified exergaming effects on enjoyment and attitude towards exergames. No interaction effects on achievement motivation, exergame intention or PEx intention were found.
Conclusions: This is the first study to examine an intervention combining Mechanics-Dynamics-Aesthetics (MDA) gamification framework as a didactic method and exergame as an educational resource, adopted as a strategy to promote the variables associated with encouraging PEx in primary schools. A gamified exergaming educational intervention might have some positive effects on variables associated with physical exercise promotion but its ability as an exercise promotion strategy is still unclear and future research should be conducted.

Introduction

Physical education (PE) has significant direct and indirect effects on students' physical exercise (PEx), especially through school intervention programmes in compulsory education (Martins et al. 2017). The systematic review by Errisuriz et al. (2018) stated the need to incorporate the measurement of psychological variables (not only physical-physiological variables) into experimental studies of school interventions to assess effects on PEx.

Physical activity has been conceived as any intentional body movement (Devís 2001). PEx is planned, structured and repetitive physical activity (Caspersen, Powerl, and Christenson 1985). Based on the latest global review studies (Chaput et al. 2020), the World Health Organization indicates that the optimal level of physical activity for children aged 10–12 years to present health benefits is approximately 60 minutes a day, or even slightly longer, and a little more, and physical activity is understood as being moderate-vigorous. Psychologically, this is a key age range because it is a critical period during which to establish or lose habits later in adolescence and adulthood. This age group is believed to be critical for its impact on psychological and behavioural risk factors (Azevedo et al. 2014), especially towards girls' physical activity (Harrington et al. 2018).

Children and young people spent a lot of time on digital games and sports (Campos 2015; Chacón et al. 2015, 2016, 2017), and this digital leisure can be associated with a sedentary lifestyle (Chacón et al. 2015). Some reports indicate that 86% of Spanish children aged 10–14 years spend at least 1 hour in front of a screen during their leisure time (INE 2017).

Exergames are digital motor games that aim to stimulate players' motor skills and are popular on the global market, to which academic research has paid increasing attention (Lin 2015; Quintas 2019). Exergames, understood as a type of game, are applied to PE and can simultaneously provide the benefits of motor games and video games (Gee 2003; González and Navarro 2015), such as the benefit of 'active learning' (an environment inviting players to act), 'psychological moratorium' (learners dare to take risks when the real consequences of their actions have a mild or little impact), 'Situated meaning' (where meanings of signs are always situated from and in personal experience), and enactive learning (motor learning by doing).

Some scientific studies propose incorporating exergames into PE classes to increase children's physical activity (Lwin and Malik 2012; Chacón et al. 2016) and to promote PEx practice outside classrooms (Li and Lwin 2016) because practicing PEx is associated with health, and physical-psychological well-being (Wu et al. 2017). Exergames imply more physical activity and, generally, more health benefits than traditional sedentary activities or sedentary video games (Staiano and Calvert 2011; Lau, Wang, and Maddison 2016). A recent review concludes that exergames can be

useful for promoting PEx and health in the adult population that does not normally perform traditional PEx, provided that duration and intensity of exergame-associated exercise recommendations are met (Street, Lacey, and Langdon 2017). However, the effect of promoting PEx by exergaming has been demonstrated only in the short term (Williams and Ayres 2020), and there is some ambiguity as to the degree to which video games (non-exergaming) are, or are not, significantly associated with obesity in children and if, in turn, there is any evidence for increased exergaming use reducing screen and video game times (Kracht, Joseph, and Staiano 2020). This may imply that the use of exergaming can indirectly favour children's health as it can psychologically promote PEx practice.

Exergames, gamification and psychological PEx promotion

Several exergame-based interventions have been implemented as a psychological strategy; that is, as a strategy to change attitudes, habits and behaviours associated with PEx (Li and Lwin 2016; Lwin et al. 2016; Nguyen et al. 2016; Robertson et al. 2016; Watson et al. 2016). Less is known about exergames' ability to promote PEx in primary schools (Li and Lwin 2016). Studies defend the need to explore the psychological effects of interventions based on both exergames and PE (Lee et al. 2017) because it is still not known how it affects school performance (Bonde et al. 2014), motivation in class (Bronner, Pinsker, and Adam Noah 2015) and if it can really promote habits outside the classroom for activities not based on exergame (Garn et al. 2012; Lwin and Malik 2012; Nguyen et al. 2016) as this could allow its possible virtues to be known in education and healthy terms.

The analytical study of a technological tool in education makes didactic sense if it is associated with a formative intervention framework (Rivoltella 2017, 62). Therefore, introducing exergames as a tool is not enough to bring about an essential change, and would be necessary to add an educational strategy as gamification to change the dynamics of lessons. As Dichev and Dicheva (2017) established, besides exergames being a technological resource, it is necessary to study gamification as a new didactic method in usual PE classes. Following these same authors, gamification refers to the use of game-based elements in non-game contexts to motivate action by making activities more game-like.

It has been specifically suggested that exergames should be incorporated into PE classes to enhance students' motivation (Campelo et al. 2015). Several motivation-related psychological exergaming benefits have already been proven in PE classes, such as favouring change in physically active behaviour (Lwin and Malik 2012) or increasing situational interest (Sun 2013). Gamification has also been associated with motivation to learn in educational contexts (Dichev and Dicheva 2017). However, gamification can act as a double-edged sword for motivation, depending on its design (Peng et al. 2012; Hanus and Fox 2015; Gu and Solmon 2016; Ryan and Deci 2017).

Exergames, gamification and PEx promotion through achievement motivation and enjoyment

The Achievement Goal Theory (AGT) (Nicholls 1984) has proven useful for explaining students' behaviour, cognition and affect in PE and PEx outdoors (Gu and Solmon 2016). The AGT conceives the individual as an intentional organism, directed towards a goal that rationally functions. An individual's goals consist of striving to demonstrate competence and ability in achievement contexts, which can be mastery-oriented or performance-oriented. Thus perceived motor competence (PMC) is a fundamental motivational variable. Achievement is interpreted subjectively by the individual, as are success and failure. Environments created by educational agents (teachers, parents, etc.) in achievement contexts have been called 'motivational climates' (Nicholls 1984). The educational environment is an achievement context that can influence the orientation of an individual's goals. According to motivational theories of achievement, Nishida (1988) built a transcultural model applied to PE, and discovered that schoolchildren who were more motivated to mastery

better performed academically during PE than those who manifested motivation aimed to avoid. Nishida's achievement motivation model includes three confirmed constructs: engagement and dedication in learning, PMC, anxiety of error and stress situations (Nishida 1988; Graupera et al. 2004). Positive relations appear between PMC in motivation and intentions to be physically active in the primary PE field (Ntoumanis and Standage 2009; Franco and Coteron 2017; van Aart et al. 2017; Márquez, Azofeifa, and Rodríguez 2019). Peng et al. (2012) stated that exergaming can increase players' perceived competence and efficacy, and can be further enhanced by showing achievements. In gamification, the dynamic adjustment of difficulty levels and badges led to increased achievement for motivation-related dimensions (Peng et al. 2012). Badges, leaderboards and performance graphs also positively affect achievement motivation and perceived task meaningfulness (Sailer et al. 2017). Conversely to these findings, Mekler et al. (2017) observed how points, leaderboards and levels did not affect achievement motivation-related dimensions, but performance quantity (i.e. more time spent on educational tasks). They argued that these game design elements can function as extrinsic incentives. However according to Sailer et al. (2017), this is considered a more positive tendency than a negative one for the possibility of deliberately influencing achievement motivation with gamification.

Enjoyment, referred to as the level of excitement or pleasure that a person derives from playing (Li and Lwin 2016), forms an essential part of any game because it is an important factor to predict future play times and play duration (Limperos and Schmierbach 2016). Enjoyment has been a commonly cited reason for playing exergames (Lee et al. 2017). It can be implemented by either feedback, challenge and rewards (Lyons 2015) or mechanisms shared by exergames and gamification. Exergames can increase physical activities through enjoyment (Ho et al. 2017).

Attitude towards exergames and PEx promotion

Exergames come over as a very promising option to achieve an active life, especially with wearable fitness technologies (Beltran-Carrillo et al. 2015). PEx through exergaming is another form of PEx. Increasing intention to play exergames in the future (exergame intention) is a way to increase intention to practice PEx. Attitude towards exergames predicts exergame intention (Kari and Makkonen 2014), which suggests that attitude towards exergaming plays an important role in individuals' future intentions to play exergames. Positive effects on exergame intention have been found in intervention studies based on exergames (Song, Peng, and Lee 2011). Moreover, exergame intention positively correlates with attitude towards exergames (Ho et al. 2017). Exergame intention is also positively related to the future intention of doing PEx (PEx intention) (Garn et al. 2012; Li and Lwin 2016). A study conducted with university students found that PEx intention was greater for exergaming interventions than for generic PEx (Garn et al. 2012). Lwin and Malik (2012) reported improved PEx intention with an exergaming-based intervention. Another educational intervention based on exergaming indicated participants' good attitude towards exergames, although this positive attitude did not significantly change over time (Keskinen et al. 2014). Other studies obtained no significant results for attitudes towards exergames and their future use (Lee et al. 2017).

Objective and hypotheses

Empirical support for the effectiveness of exergaming on students' motivation and in-class activity in PE is sparse (Sun 2013). Several studies on the attitude towards exergames and their future use mostly found no significant results (Lee et al. 2017), others have reported no changes after interventions (Quinn 2013; Keskinen et al. 2014), and some even report exergames failing to promote physical activity (Madsen et al. 2007; Lau, Wang, and Maddison 2016). Studies are needed to advance the body of knowledge not only on the physical-physiological effects that have already been widely studied, but also the psychological effects of exergame (Gao, Zhang, and Stodden 2013; Azevedo et al. 2014; Vernadakis et al. 2014; Street, Lacey, and Langdon 2017). Moreover, recent studies

indicate the limitations of empirical research on gamified education, such as the dominant application to colleges (Dichev and Dicheva 2017) or lack of comparison groups (Hanus and Fox 2015). All these reasons support the need to design a natural experiment based on gamification principles (Hunicke, LeBlanc, and Zubek 2004), and a recent exergame compatible with the school PE curriculum to be applied to a relevant age group (10–12 years) at several primary schools. In fact, our two previous studies showed that exergame and gamification may be useful and applicable for didactics in Physical Education at school, producing positive psychological effects on basic psychological needs, flow-related dimensions and academic performance (Quintas et al. 2020; Quintas-Hijós, Peñarrubia-Lozano, and Bustamante 2020). However, as far as we know, no research has used a natural experimental method with a control and experimental intervention applied by the same teacher to analyse the promotion of PEx with gamification and exergaming in schoolchildren. Thus, based on the same natural experiment from our two previous studies, the aim of this study was to analyse the effects of a gamified exergaming intervention in 10–12 year-old Primary Education students on the psychological variables associated with promoting PEx and relevant to PE, such as achievement motivation, enjoyment, attitude towards exergames, exergame intention and PEx intention (Martins et al. 2017).

Finally, as the intervention of this study was based on the limitations and suggestions of other empirical studies, the hypotheses (H) were postulated in line with the results of previous studies (Ntoumanis and Standage 2009; Fernandez et al. 2011; Song, Peng, and Lee 2011; Garn et al. 2012; Peng et al. 2012; Kari and Makkonen 2014; Keskinen et al. 2014; Lwin and Malik 2014; Li and Lwin 2016; Franco and Coteron 2017; Lee et al. 2017; Mekler et al. 2017; van Aart et al. 2017; Márquez, Azofeifa, and Rodríguez 2019). Hence the gamified exergaming intervention will produce more achievement motivation (H1), more enjoyment (H2), a more positive attitude towards exergames (H3), more exergame intention (H4) and more PEx intention (H5) in students than non-gamified and non-exergaming intervention with time.

Material and method

Sample

The participants were the same as those in our previous study (Quintas et al. 2020). The sample comprised 417 students (53.2% girls, $n = 222$; 46.8%, $n = 195$) from four Primary Education schools (two public, two non-public). Their mean age was 11.1 (SD = 1.7), and 50.4% of the sample studied Primary Education Year 6 (aged 10–11 years; $n = 210$) and 49.6% studied Primary Education Year 7 (11–12 years; $n = 207$). The study was carried out in 22 groups-classes from the four schools (minimum ratio of children = 15, maximum = 25).

The criteria to select schools were: their material adequacy (facilities, Wi-Fi connectivity), public/private schools' diversity, their students' ethnic and socio-economic diversity, schools from different cities, teaching staff's positive predisposition and accessibility for researchers. All the Year-6 and Year-7 students were invited to participate in the study by signed parental consent ($n = 418$). As one student did not provide consent and was unable to participate in the study, no data about him were collected, although he went to class as usual. The study received the ethical approval of the Ethical Committee of Clinical Research of Aragón (Spain) (statement no.: 10/2017).

A chi-square analysis found no differences in distribution by sex χ^2 (1, $N = 417$) = .018, $p = .893$. The independent samples t-test revealed no significant differences in attendance ($t(381) = -1.065$, $p = 0.287$).

Study design

Like in our previous study (Quintas et al. 2020), a natural experiment with a non-randomised controlled design was conducted with a pre- and a post-measure. The study applied quasi-experimental

design. This design allows maximum control without losing the naturalness of school cohorts and it is considered appropriate for studies of a similar nature (Verjans-Janssen et al. 2018).

The experimental treatment was designed (gamified exergaming intervention; Figure 1), based on usual didactic dance teaching in Spanish PE (Larraz 2012), but added a gamified climate in class and used an exergame. Another treatment was designed, the control one (traditional didactic intervention; Figure 2), which was similar to the experimental treatment, but without either the gamified climate in class or the exergame. Before receiving either educational treatment, each student personally completed the digital questionnaires using computers or tablets during a school session. Each treatment lasted 12 sessions, or 9 h, and was applied at each school for 4 weeks during curricular PE classes. Treatments were put into practice with the Year-6 and Year-7 students. Both the experimental treatment and control were applied at each school, but each student received only one treatment, the experimental or the control, depending on whether (s)he belonged to Year 6 or Year 7. The school year (Year 6 or Year 7) at which each intervention was implemented was randomly decided. For ethical and methodological reasons, it was decided that all the classes-groups from the same school and Year would receive the same treatment. At the end of the 4-week programme, the participants filled in the questionnaires under the same conditions. Data collection took place between September 2017 and July 2018.

The traditional didactic treatment and the gamified exergaming treatment were applied in the same way by the same teacher-researcher at all four schools in the different cities for 10 months; that is, the same teacher taught the entire study sample to, thus, reduce any intervention bias stemming from involving several teachers. The teacher was not their usual teacher at each school, who only taught the classes for this study.

Materials

The gamification system based on Points-Badges-Leaderboards is the most widely applied and well-studied. However, it only matches one player type: 'achievers' (Bartle 2003). Therefore, as in our previous study (Quintas et al. 2020), the Mechanics-Dynamics-Aesthetics (MDA) framework by Hunicke, LeBlanc, and Zubek (2004) was used to achieve mastery-oriented motivational climates and an inclusive gamified atmosphere for all types of student-players. Mechanics refers to the

Figure 1. Experimental group dancing level 1 'Rasputin' with smartphones held in hands.

Figure 2. Control group dancing salsa without either an exergame or gamification.

system's set of constituent elements, the relation linking them, and the way in which a system can routinely function. Dynamics refers to the way in which mechanics effectively works; that is, how the player-student interacts with the mechanics. Aesthetics refers to both the perceptions produced by the mechanics in the player-student interaction as it is designed (beauty) and the sensations-emotions felt by students while playing (Quintas 2020). The MDA framework was adapted to the education field, and all the applied elements are specified in Table 2. No negative points of any kind were used. Mechanic elements were designed: a system of positive points, rewards, classifications, levels of difficulty, challenges, achievements, badges, cooperative and competitive teams, virtual avatars, and the possibility of personalising avatars. Self-referential situations with an epic cooperative group sense were used to promote motivation in all the student types, and even with those who were not competitive.

Table 1. The applied MDA framework elements (extracted from Quintas et al. (2020)).

Mechanics	Dynamics	Aesthetics
Dance performance point	Reinforcement Cumulability	Pleasure, Satisfaction, absortion
Creativity point		
Attention point		Social membership
Good behaviour point		
Design a choreography	Self-expression	Pleasure, Identity
Design a group choreography	Cooperation	Social membership
Leaderboard	Competition	Social membership, Identity
Star badge (perfect dance)	Reinforcement Progress Collectability	Satisfaction
Green badge (for individual improvement).		Fun, Pleasure, Interest
Group Green Badge	Cooperation Collectability	Social membership, Identity
Group improvement point		
Best Dancers Badges	Competition	
Badge dancers that have improved the most		
Point helping the team	Cooperation, Collectability	
Point of dance plank	Status Competition	Identity, absortion
Custom avatar	Self-expression	Customise, Identity, Beauty
Increasingly difficult dance levels	Progress	Fun, Satisfaction, Interest
Choosing dance Level 9 from more than 300 dances.	Self-expression	Identity, Customise
Just Dance Now and ClassDojo interface	Progress	Beauty, absortion
Music in all classes	Self-expression	Interest, Pleasure

Table 2. Descriptive data in the variables of interest: mean and standard deviation (sample size).

	Control treatment		Experimental treatment	
	Baseline	Follow-up	Baseline	Follow-up
Attendance	–	9.61 ± .79 (191)	–	9.69 ± .69 (226)
Achievement motivation	114.76 ± 15.86 (159)	110.73 ± 14.09 (159)	115.22 ± 15.37 (199)	114.02 ± 15.8 (199)
Engagement and delivery in learning	60.52 ± 8.59 (171)	58.15 ± 9.66 (171)	60.49 ± 7.9 (205)	59.21 ± 8.28 (205)
Perceived motor competence	19.59 ± 5.81 (178)	19.2 ± 5.9 (178)	20.25 ± 6.08 (214)	20.42 ± 5.65 (214)
Anxiety in the face of error and stress situations	33.98 ± 12.23 (172)	33.67 ± 12.58 (172)	34.89 ± 12.69 (208)	35.02 ± 13.56 (208)
Satisfaction towards PE	27.74 ± 2.89 (178)	26.97 ± 3.54 (178)	27.35 ± 3.39 (218)	27.41 ± 3.34 (218)
Enjoyment in PE	22.76 ± 3.05 (166)*	22.13 ± 3.22 (166)	21.85 ± 3.71 (211)*	22.22 ± 3.18 (211)
Boredom in PE	3.1 ± 1.94 (174)	3.34 ± 1.85 (174)	3.41 ± 2.01 (215)	3.26 ± 1.78 (215)
Attitude toward exergames	22.63 ± 6.18 (177)	21.79 ± 7.10 (177)	23.55 ± 5.2 (211)	24.93 ± 4.43 (211)
Exergame intention	14.55 ± 4.87 (177)	14.36 ± 5.43 (177)	16.01 ± 4.02 (209)	16.66 ± 3.51 (209)
Physical exercise intention	15.77 ± 4.38 (177)	15.72 ± 4.53 (177)	15.38 ± 4.21 (216)	15.75 ± 3.97 (216)

* Significant differences between groups at the baseline were found ($t(374) = 2.613$, $p = .009$).

In order to facilitate the application of many MDA framework elements (Table 1), the ClassDojo app was used because it is one of the most widespread technological applications (Torres 2017). This application allows evaluations or measurements for a scientific purpose, it is suitable for daily teaching, it has a high aesthetic component and it is related to Primary Education (Casanova and Serrano 2019). Each student's points accumulated in the ClassDojo app and could be consulted by students at any time, even at home.

In the experimental group, the Just Dance Now exergame web platform was employed (https://justdancenow.com/). Students could dance in groups in front of a large projected screen while they held a Samsung J3 smartphone in their right hand (see Figure 1). As smartphones had an accelerometer that detected movement, each smartphone detected every student's dance points. The control group learned to dance without using any gamifier resource or exergame (see Figure 2). The Just Dance Now exergame was employed because it is compatible with school facilities and is based on accessible materials (screen projector, laptop, smartphone, the Internet). It use is also justified by its great commercial success worldwide, its marked usage in young people's leisure time (Allsop et al. 2013), and because it has been scientifically studied (Lin 2015; Gao et al. 2016; Li and Lwin 2016). The experimental treatment design allowed all the students to dance several times during all the sessions.

Measures

Achievement motivation. Students filled in the Achievement Motivation in Physical Education Test (AMPET) (Nishida 1988), validated and adapted to the Spanish population (Graupera et al. 2004), which comprises 37 items. Items were measured on a 5-point Likert scale from 1 (Strongly Disagree) to 5 (Strongly Agree; the Cronbach α applied to our sample was = .84). This scale measured achievement motivation with three dimensions: engagement and dedication in learning ($\alpha = .88$), PMC ($\alpha = .81$), and anxiety before error and stress situations ($\alpha = .91$) (see the Appendix).

Enjoyment. The participants completed the Sport Satisfaction Instrument (SSI), validated and adapted to PE and the Spanish population by Baena et al. (2012), which contains eight items. Items were measured on a 5-point Likert scale from 1 (Strongly Disagree) to 5 (Strongly Agree). This instrument measures two independent constructs: enjoyment ($\alpha = .81$) and boredom ($\alpha = .54$). To achieve acceptable reliability for 'boredom' ($\alpha = .71$), the second item was eliminated ('*In Physical Education classes, I often daydream instead of thinking about what I'm really doing, already warned by them*'), as pointed out by (Baena et al. 2012) (see the Appendix).

Attitude towards exergames. Students filled in the 'Attitudes Toward Exergames' scale, created and validated by Ho et al. (2017) with children aged 9–12 years. It is translated into Spanish and

replaces 'X-Box kinect' with a 'video game in which I have to move my whole body'. This instrument is composed of four items, and measures attitudes towards exergames on a 5-point Likert scale from 1 (Strongly Disagree) to 5 (Strongly Agree; $\alpha = .95$) (see the Appendix).

Exergame intention. Participants filled in the 'Exergame intention' scale based on Li and Lwin (2016) by replacing 'the game' with 'a video game in which I have to move my whole body'. Four questions were asked and answers were selected on a five-point scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree; $\alpha = .91$) (see the Appendix).

PEx intention. Students completed the 'Exercise Intention' scale created by Li and Lwin (2016). This instrument comprises four items. It measures exercise intention on a 5-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree; $\alpha = .91$) (see the Appendix).

Statistical analyses

Given our previous study (Quintas et al. 2020) we conducted the data analyses described below. The baseline differences between groups were compared by an independent *t*-test for the normally distributed variables and the Mann–Whitney U test for the abnormally distributed variables (see Table 2). To test the research H, the two study groups were compared using factorial ANOVAs 2 (time; pre-treatment condition vs. post-treatment condition) × 2 (treatment; traditional didactic intervention vs. gamified exergaming intervention). One-way ANOVAs were also applied to clarify the obtained effects and the interpretative process. The effect size was calculated by eta partial eta-squared (η_p^2), and the analysis threshold was set at ≤0.0125 with Bonferroni adjustment. This adjustment is adequate to gain validity with such a design and the statistical analysis of multiple comparisons because it is a more restrictive criterion that reduces the probability of committing type I errors; that is, a false-positive (Bland and Altman 1995). All the statistical analyses were performed using SPSS (version 22.0, https://www.ibm.com/es-es/analytics/spss-statistics-software).

Results

The means and standard deviations of all the variables of treatments can be found in Table 2. The H testing is shown below.

H1 predicted that, with time, the gamified exergaming education intervention would produce more achievement motivation in students than the non-gamified and non-exergaming intervention. No interaction effect (Time × Treatment) was found ($p > .0125$). However, a time effect on achievement motivation ($F(1) = 11.83$, $p = .001$, $\eta_p^2 = .032$) was noted. Performing a more specific analysis in each group by a one-way ANOVA (Post–Pre) showed that the control group displayed less achievement motivation with time ($F(1) = 12.19$, $p = .001$, $\eta_p^2 = .072$). However, the experimental group did not significantly display more or less achievement motivation with time ($F(1) = 1.45$, $p = .229$, $\eta_p^2 = .007$). Thus H1 was not supported.

H2 predicted that, with time, the gamified exergaming education intervention would produce more enjoyment in students than the non-gamified and non-exergaming intervention. An interaction effect was found on enjoyment ($F(1) = 7.56$, $p = .006$, $\eta_p^2 = .02$) (see Figure 3). Thus H2 was supported.

H3 predicted that, with time, the gamified exergaming education intervention would produce a more positive attitude towards exergames in students than the non-gamified and non-exergaming intervention. An interaction effect ($F(1) = 7.08$, $p = .008$, $\eta_p^2 = .018$, see Figure 4) and a treatment effect ($F(1) = 24.64$, $p = .000$, $\eta_p^2 = .06$) on attitude towards exergames took place. Thus H3 was supported.

H4 predicted that, with time, the gamified exergaming education intervention would produce more exergame intention in students than the non-gamified and non-exergaming intervention. No interaction effect (Time × Treatment) was found ($p > .0125$), but a treatment effect on exergame intention was noted ($F(1) = 22.94$, $p = .000$, $\eta_p^2 = .056$). Performing a more specific analysis in each

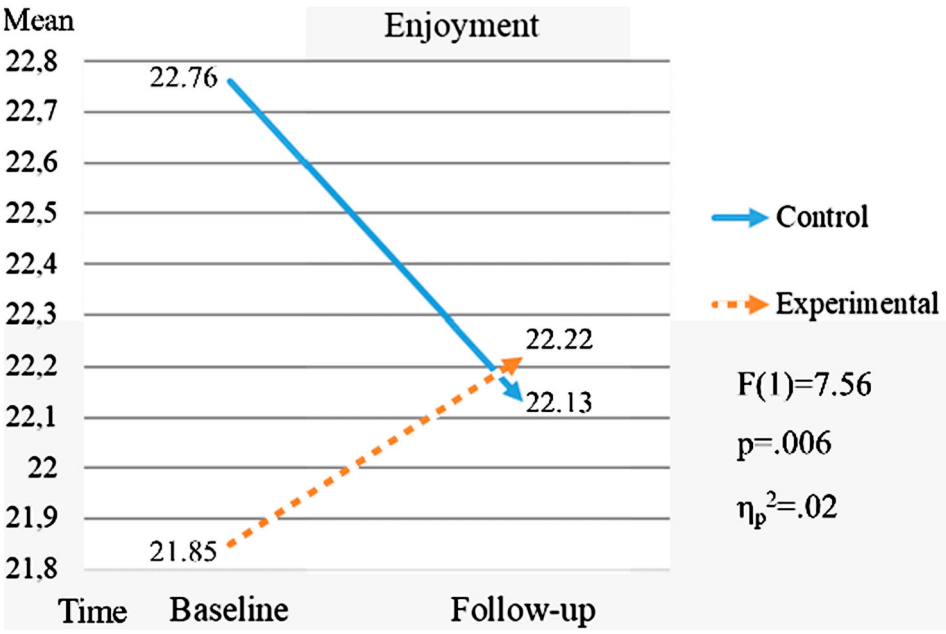

Figure 3. Interaction effect on enjoyment.

group by a one-way ANOVA (Post–Pre) showed that the experimental group's exergame intention significantly improved with time ($F(1) = 4.8$, $p = .03$, $\eta_p^2 = .023$), which was not observed with the control group. Thus H4 was not supported.

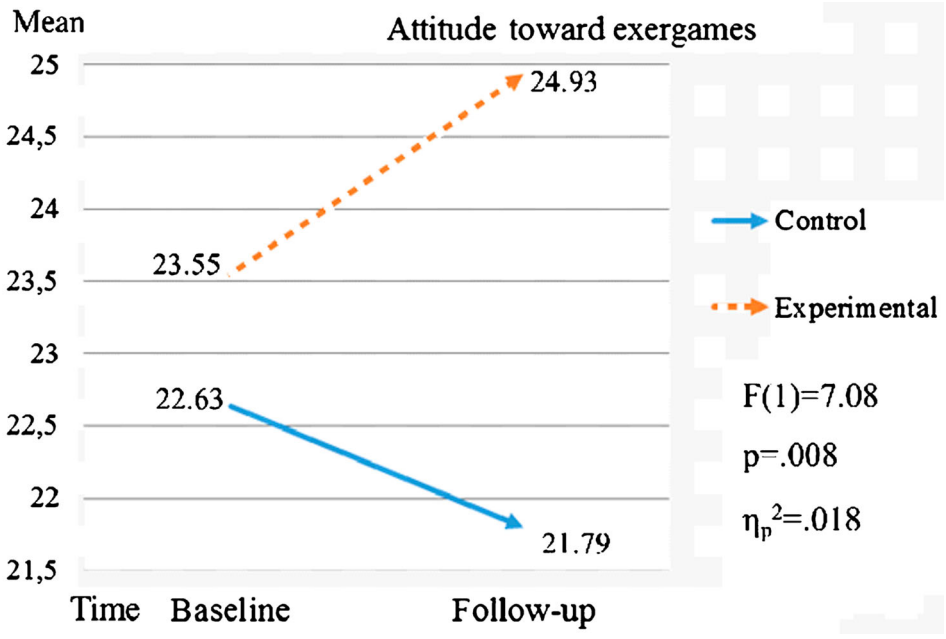

Figure 4. Interaction effect on attitude toward exergames.

H5 predicted that, with time, the gamified exergaming education intervention would produce a more PEx intention in students than the non-gamified and non-exergaming intervention. No interaction effect or main effects were found on PEx intention ($p > .0125$). Thus H5 was not supported.

Discussion

The aim of this study was to analyse the effects of a gamified exergaming school intervention on the psychological variables associated with PEx promotion. The hypothesis test allowed us to affirm that a gamified exergaming educational intervention, compared to another almost identical intervention, but with neither gamification nor exergames, could have some positive effects on the variables associated with promoting PEx, like enjoyment or attitude towards exergames. That is, the experimental intervention would apparently have more benefits than the control intervention for some variables associated with promoting PEx, but not directly for exercise intention.

The results of the present study on achievement motivation suggest a differential effect of the gamified exergaming educational intervention compared to the non-gamified and non-exergaming intervention. Although no interaction effect was obtained, the results revealed a lower achievement motivation level in the control group with time, which was not observed in the experimental group. This effect could be related to increased engagement and delivery in learning (Sun 2013; Bronner, Pinsker, and Adam Noah 2015). The experimental group's engagement might be associated with a higher cognitive ability or more involved physical effort (Best 2012; Bronner, Pinsker, and Adam Noah 2015).

The experimental group obtained more enjoyable experience than the control group, which coincides with Garn et al. (2012). The explanation for this improvement probably stems from the presence of continuous feedback, challenge and rewards (Lyons 2015), which students did not interpret as controlling factors (Hanus and Fox 2015), and are essential in both exergame and gamification. Enjoyment is the feature of the current gamified intervention that would most promote PEx and the future use of exergames because it is a significant predictor of both play duration and the number of times for play to occur (Limperos and Schmierbach 2016).

The best beneficial effects obtained in the experimental treatment on attitude towards exergames coincided with not only the findings of some studies with a similar study design (Garn et al. 2012), but also with studies that included a different methodological control (Keskinen et al. 2014; Campelo et al. 2015). This attitudinal improvement may be due to greater lived enjoyment (Li and Lwin 2016). However, more positive results were expected for exergame intention (Lin 2015) and PEx intention, despite a treatment effect having been found on exergame intention. These results contrast with the study, which found that playing exegames could improve participants' attitudes towards and intention to do other forms of PEx (Garn et al. 2012; Lwin and Malik 2012; Nguyen et al. 2016). These differences may be due to the selection of sedentary or obese populations (Garn et al. 2012), not using official PE subject hours (Lwin and Malik 2012; Nguyen et al. 2016) or participants were older (Nguyen et al. 2016). Besides, it is possible that no mastery-oriented motivational climate was built. According to the AGT (Nicholls 1984), this would explain lack of improvement in achievement motivation-related dimensions like PMC and, therefore, lack of PEx intention. Other studies have reported that exergames failed to promote PEx because participants considered them boring (Lau, Wang, and Maddison 2016), but we did not find this reason. The ability of both exergame and gamification to produce PEx remains unclear, and more studies need to be done.

This study has several limitations. One-month interventions (9 h) were carried out given the traditional way of scheduling school contents. However, longitudinal studies are necessary to ascertain the long-term psychological effects. Another study limitation is its didactic-scientific design that included no single differentiating factor between the experimental and control groups because the experimental group engaged in both a gamified and, in parallel, an exergaming intervention. However, both treatments were equivalents for the other didactic elements. Future research

could propose studies that specifically and separately analyae the effects of these two factors. Given our study design, the obtained results are generalisable to contexts based on gamification and exergame at school ages. However, the gender effect may have been important because cultural dance associations with girls exist in Spain (Azevedo et al. 2014), as do some video games associations exist with boys (Díez-Gutiérrez et al. 2004). Thus it would be very interesting to analyse gamification and exergame in contents other than dance, and to also consider other PE-related variables, such as the actual amount of exercise performed in non-education contexts.

By way of conclusion, this study is the first to examine an intervention that combines MDA gamification as a didactic method and exergame as an educational resource, adopted as a strategy to promote the variables associated with encouraging PEx in primary schools. A rigorous scientific design in schools was implemented, and groups were compared, along with reliable measures such as scientific literature requests. This study evidences the possibility of joining methodological control techniques and the good naturalness of the education context. This is the only empirical study based on a natural experiment that has investigated educational gamification and exergaming with significant results. The discovered positive effects on enjoyment and attitude towards exergames make this study one of the very few to provide positive evidence for educational gamification. Improved enjoyment suggests that gamified design generates a more task-oriented motivational climate, but no conclusive results for the PEx intention variable were obtained. The ability of gamification and exergame as an exercise promotion strategy is not yet clear and future research should be conducted.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by Fundación Bancaria Ibercaja [Grant Number JIUZ-2017-SOC-06]; FEDER (Programa Operativo 2014-2020) and the Regional Aragonese Government [Grant Number INFR2016_UZ_SOC_05]; Fundación Hergar [Grant Number 2018/0081].

ORCID

Alejandro Quintas http://orcid.org/0000-0003-4621-6993
Juan-Carlos Bustamante http://orcid.org/0000-0002-8201-0469

References

Allsop, S., P. L. S. Rumbold, D. Debuse, and C. Dodd-Reynolds. 2013. "Real Life Active Gaming Practices of 7-11-Year-Old Children." *Games for Health Journal* 2 (6): 347–353. doi:10.1089/g4h.2013.0050.

Azevedo, L. B., D. Burges Watson, C. Haighton, and J. Adams. 2014. "The Effect of Dance Mat Exergaming Systems on Physical Activity and Health – Related Outcomes in Secondary Schools: Results from a Natural Experiment." *BMC Public Health* 14 (1): 951. doi:10.1186/1471-2458-14-951.

Baena, A., A. Granero-Gallegos, C. Bracho-Amador, and F. J. Pérez-Quero. 2012. "Spanish Version of the Sport Satisfaction Instrument (SSI) Adapted to Physical Education." *Revista de Psicodidáctica* 17 (2): 377–395.

Bartle, R. 2003. *Designing Virtual Worlds*. Berkeley, CA: New Riders.

Beltran-Carrillo, V. J., J. I. Beltran-Carrillo, D. Gonzalez-Cutre, S. J. H. Biddle, and C. Montero-Carretero. 2015. "Are Active Video Games Associated With Less Screen Media or Conventional Physical Activity?" *Games and Culture* 11 (6): 608–624. doi:10.1177/1555412015574941.

Best, J. R. 2012. "Exergaming Immediately Enhances Children's Executive Function." *Developmental Psychology* 48 (5): 1501–1510. doi:10.1037/a0026648.

Bland, J. M., and D. G. Altman. 1995. "Multiple Significance Tests: The Bonferroni Method." *British Medical Journal* 310: 170.

Bonde, M. T., G. Makransky, J. Wandall, M. V. Larsen, M. Morsing, H. Jarmer, and M. O. A. Sommer. 2014. "Improving Biotech Education Through Gamified Laboratory Simulations." *Nature Biotechnology* 32: 694–697. doi:10.1038/nbt.2955.

Bronner, S., R. Pinsker, and J. Adam Noah. 2015. "Physiological and Psychophysiological Responses in Experienced Players While Playing Different Dance Exer-Games." *Computers in Human Behavior* 51: 34–41. doi:10.1016/j.chb.2015.04.047.

Campelo, A. M., G. Donaldson, D. P. Sheehan, and L. Katz. 2015. "Attitudes Towards Physical Activity and Perceived Exertion in Three Different Multitask Cybercycle Navigational Environments." *Procedia Engineering* 112: 256–261. doi:10.1016/j.proeng.2015.07.238.

Campos, J. A. 2015. *El uso de las TIC, dispositivos móviles, y redes sociales en un aula de la educación secundaria obligatoria*. Granada: University of Granada.

Casanova, O., and R. Serrano. 2019. "Herramientas tecnológicas asociadas a la gamificación en la formación inicial docente musical de secundaria." Paper presented at the VIII Multidisciplinary International Conference on Educational Research. Education: The Door to Any Social Improvements, Lérida.

Caspersen, C. J., K. E. Powerl, and G. M. Christenson. 1985. "Physical Activity, Exercise, and Physical Fitness: Definitions and Distinctions for Health-Related Research." *Public Health Reports* 100 (2): 126–131.

Chacón, R., M. Castro, F. Zurita, T. Espejo, and A. Martínez. 2016. "Videojuegos Activos como recurso TIC en el Aula de Educación Física: estudio a partir de parámetros de Ocio Digital." *Digital Education Review* 29: 112–123.

Chacón, R., T. Espejo, A. Cabrera, M. L. Castro, J. F. López, and F. Zurita. 2015. "«Exergames» para la mejora de la salud en niños y niñas en edad escolar: estudio a partir de hábitos sedentarios e índices de obesidad." *RELATEC: Revista Latinoamericana de Tecnología Educativa* 14 (2): 39–50. doi:10.17398/1695-288x.14.2.39.

Chacón, R., F. Zurita, A. Martínez, M. Castro, T. Espejo, and C. Pinel. 2017. "Relación entre factores académicos y consumo de videojuegos en universitarios. Un modelo de regresión." *Píxel-Bit. Revista de Medios y Educación* 50: 109–121. doi:10.12795/pixelbit.2017.i50.07.

Chaput, J. P., J. Willumsen, F. Bull, R. Chou, E. Ekelund, J. Firth, R. Jago, F. B. Ortega, and P. T. Katzmarzyk. 2020. "2020 WHO Guidelines on Physical Activity and Sedentary Behaviour for Children and Adolescents Aged 5-17 Years: Summary of the Evidence." *The International Journal of Behavioral nutrition and Physical Activity* 17 (1): 141. doi:10.1186/s12966-020-01037-z.

Devís, J. 2001. *La educación física, el deporte y la salud en el siglo XXI*. Alcoy: Marfil.

Díez-Gutiérrez, E. J., E. Terrón Bañuelos, M. García-Gordón, J. Rojo, R. Cano González, R. Castro Fonseca, R. E. Valle Flórez, O. Fontal Merillas, and J. D. Morala. 2004. *La diferencia sexual en el análisis de los videojuegos*. Madrid: CIDE: Instituto de la Mujer.

Dichev, C., and D. Dicheva. 2017. "Gamifying Education: What Is Known, What Is Believed and What Remains Uncertain: A Critical Review." *International Journal of Educational Technology in Higher Education* 14 (1): 9. doi:10.1186/S41239-017-0042-5.

Errisuriz, V., N. Golaszewski, K. Born, and J. Bartholomew. 2018. "Systematic Review of Physical Education-Based Physical Activity Interventions Among Elementary School Children." *The Journal of Primary Prevention* 39 (3): 303–327. doi:10.1007/s10935-018-0507-x.

Fernandez, J., A. Méndez, J. Cecchini, and C. González. 2011. Achievement Goals and Social Goals' Influence on Physical Education Students' Fair Play // La influencia de las metas de logro y las metas sociales sobre el fair play de estudiantes de Educación Física de Secundaria (Vol. 17).

Franco, E., and J. Coteron. 2017. "The Effects of a Physical Education Intervention to Support the Satisfaction of Basic Psychological Needs on the Motivation and Intentions to Be Physically Active." *Journal of Human Kinetics* 59 (1): 5–15. doi:10.1515/hukin-2017-0143.

Gao, Z., J. E. Lee, Z. Pope, and D. Zhang. 2016. "Effect of Active Videogames on Underserved Children's Classroom Behaviors, Effort, and Fitness." *Games for Health Journal* 5 (5): 318–324. doi:10.1089/g4h.2016.0049.

Gao, Z., T. Zhang, and D. Stodden. 2013. "Children's Physical Activity Levels and Psychological Correlates in Interactive Dance Versus Aerobic Dance." *Journal of Sport and Health Science* 2 (3): 146–151. doi:10.1016/j.jshs.2013.01.005.

Garn, A. C., B. L. Baker, E. K. Beasley, and M. A. Solmon. 2012. "What Are the Benefits of a Commercial Exergaming Platform for College Students? Examining Physical Activity, Enjoyment, and Future Intentions." *Journal of Physical Activity and Health* 9 (2): 311–318.

Gee, J. P. 2003. *What Video Games Have to Teach Us about Learning and Literacy*. New York: Palgrave Macmillan.

González, C., and V. Navarro. 2015. "A Structural Theoretical Framework Based on Motor Play to Categorize and Analyze Active Video Games." *Games and Culture* 11 (7-8): 690–719. doi:10.1177/1555412015576613.

Graupera, J. L., M. Gutiérrez, T. Nishida, and L. M. Ruiz. 2004. "El Test Ampet de motivación de logro para el aprendizaje en educación física: desarrollo y análisis factorial de la versión española." *Revista de educación* 335: 195–214.

Gu, X. L., and M. A. Solmon. 2016. "Motivational Processes in Children's Physical Activity and Health-Related Quality of Life." *Physical Education and Sport Pedagogy* 21 (4): 407–424. doi:10.1080/17408989.2015.1017456.

Hanus, M. D., and J. Fox. 2015. "Assessing the Effects of Gamification in the Classroom: A Longitudinal Study on Intrinsic Motivation, Social Comparison, Satisfaction, Effort, and Academic Performance." *Computers & Education* 80: 152–161. doi:10.1016/j.compedu.2014.08.019.

Harrington, D. M., M. J. Davies, D. H. Bodicoat, J. M. Charles, Y. V. Chudasama, T. Gorely, K. Khunti, et al. 2018. "Effectiveness of the 'Girls Active' School-Based Physical Activity Programme: A Cluster Randomised Controlled Trial." *International Journal of Behavioral Nutrition and Physical Activity* 15 (1): 40. doi:10.1186/s12966-018-0664-6.

Ho, S. S., M. O. Lwin, J. R. H. Song, and A. Z. H. Yee. 2017. "Escaping Through Exergames: Presence, Enjoyment, and Mood Experience in Predicting Children's Attitude Toward Exergames." *Computers in Human Behavior* 72: 381–389. doi:10.1016/j.chb.2017.03.001.

Hunicke, R., M. LeBlanc, and R. Zubek. 2004. "MDA: A Formal Approach to Game Design and Game Research." Paper presented at the Workshop on Challenges in Game AI.

INE. 2017. *Encuesta nacional de salud 2017*. Madrid: Ministerio de Sanidad, Consumo y Bienestar Social de España. https://bit.ly/3bdZyHj.

Kari, T., and M. Makkonen. 2014. "Explaining the Usage Intentions of Exergame." Paper presented at the Thirty Fifth International Conference on Information Systems, Auckland.

Keskinen, T., J. Hakulinen, M. Turunen, T. Heimonen, A. Sand, J. Paavilainen, J. Parviainen, et al. 2014. "Schoolchildren's User Experiences on a Physical Exercise Game Utilizing Lighting and Audio." *Entertainment Computing* 5: 475–484.

Kracht, C., E. Joseph, and A. E. Staiano. 2020. "Video Games, Obesity, and Children." *Current Obesity Reports* 9 (1): 1–14.

Larraz, A. 2012. "La Expresión Corporal en la Escuela Primaria experiencias desde la Educación Física." In *La expresión corporal en la enseñanza universitaria*, edited by G. Sánchez, and J. Coterón, 179–188. Salamanca: Ediciones Universidad de Salamanca.

Lau, P. W. C., J. J. Wang, and R. Maddison. 2016. "A Randomized-Controlled Trial of School-Based Active Videogame Intervention on Chinese Children's Aerobic Fitness, Physical Activity Level, and Psychological Correlates." *Games for Health Journal* 5 (6): 405–412. doi:10.1089/g4h.2016.0057.

Lee, S., W. Kim, T. Park, and W. Peng. 2017. "The Psychological Effects of Playing Exergames: A Systematic Review." *Cyberpsychology, Behavior and Social Networking* 20 (9): 513–532. doi:10.1089/cyber.2017.0183.

Li, B. J., and M. O. Lwin. 2016. "Player See, Player Do: Testing an Exergame Motivation Model Based on the Influence of the Self Avatar." *Computers in Human Behavior* 59: 350–357. doi:10.1016/j.chb.2016.02.034.

Limperos, A. M., and M. Schmierbach. 2016. "Understanding the Relationship Between Exergame Play Experiences, Enjoyment, and Intentions for Continued Play." *Games for Health Journal* 5 (2): 100–107. doi:10.1089/g4h.2015.0042.

Lin, J. H. 2015. "'Just Dance': The Effects of Exergame Feedback and Controller Use on Physical Activity and Psychological Outcomes." *Games for Health Journal* 4 (3): 183–189. doi:10.1089/g4h.2014.0092.

Lwin, M. O., S. S. Ho, J. Younbo, T. Y. Leng, R. J. Wardoyo, and K. H. Jung. 2016. "Effects of Exergaming and Message Framing in School Environments on Physical Activity Attitudes and Intentions of Children and Adolescents." *Journal of Health Communication* 21 (9): 969–978. doi:10.1080/10810730.2016.1153759.

Lwin, M. O., and S. Malik. 2012. "The Efficacy of Exergames-Incorporated Physical Education Lessons in Influencing Drivers of Physical Activity: A Comparison of Children and Pre-adolescents." *Psychology of Sport and Exercise* 13 (6): 756–760. doi:10.1016/j.psychsport.2012.04.013.

Lwin, M. O., and S. Malik. 2014. "Can Exergames Impart Health Messages? Game Play, Framing, and Drivers of Physical Activity among Children." *Journal of Health Communication* 19 (2): 136–151. doi:10.1080/10810730.2013.798372.

Lyons, E. J. 2015. "Cultivating Engagement and Enjoyment in Exergames Using Feedback, Challenge, and Rewards." *Games For Health Journal* 4 (1): 12–18. doi:10.1089/g4h.2014.0072.

Madsen, K. A., S. Yen, L. Wlasiuk, T. B. Newman, and R. Lustig. 2007. "Feasibility of a Dance Videogame to Promote Weight Loss among Overweight Children and Adolescents." *Archives of Pediatrics and Adolescent Medicine* 161 (1): 105–107. doi:10.1001/archpedi.161.1.105-c.

Márquez, M., C. Azofeifa, and D. Rodríguez. 2019. "Factores de motivación de logro: el compromiso y entrega en el aprendizaje, la competencia motriz percibida, la ansiedad ante el error y situaciones de estrés en estudiantes de cuarto, quinto y sexto nivel escolar durante la clase de educación física." *Revista Educación* 43 (1): 61–73. doi:10.15517/revedu.v43i1.33109.

Martins, J., A. Marques, M. Peralta, A. Palmeira, and F. Costa. 2017. "Correlates of Physical Activity in Young People: A Narrative Review of Reviews. Implications for Physical Education Based on a Socio-Ecological Approach." *Retos: nuevas tendencias en educación física, deporte y recreación* 31: 292–299.

Mekler, E. D., F. Brühlmann, A. N. Tuch, and K. Opwis. 2017. "Towards Understanding the Effects of Individual Gamification Elements on Intrinsic Motivation and Performance." *Computers in Human Behavior* 71: 525–534. doi:10.1016/j.chb.2015.08.048.

Nguyen, H. V., H. C. Huang, M. K. Wong, J. Lu, W. F. Huang, and C. I. Teng. 2016. "Double-Edged Sword: The Effect of Exergaming on Other Forms of Exercise; a Randomized Controlled Trial Using the Self-Categorization Theory." *Computers in Human Behavior* 62: 590–593. doi:10.1016/j.chb.2016.04.030.

Nicholls, J. G. 1984. "Achievement Motivation: Conceptions of Ability, Subjective Experience, Task Choice and Performance." *Psychological Review* 16: 328–346.

Nishida, T. 1988. "Reliability and Factor Structure of the Achievement Motivation in Physical Education Test." *Journal of Sport and Exercise Psychology* 10: 418–430.

Ntoumanis, N., and M. Standage. 2009. "Motivation in Physical Education Classes: A Self-Determination Theory Perspective." *School Field* 7 (2): 194–202. doi:10.1177/1477878509104324.

Peng, W., J. H. Lin, K. A. Pfeiffer, and B. Winn. 2012. "Need Satisfaction Supportive Game Features as Motivational Determinants: An Experimental Study of a Self-Determination Theory Guided Exergame." *Media Psychology* 15 (2): 175–196. doi:10.1080/15213269.2012.673850.

Quinn, M. 2013. "Introduction of Active Video Gaming into the Middle School Curriculum as a School-Based Childhood Obesity Intervention." *Journal of Pediatric Health Care* 27 (1): 3–12. doi:10.1016/j.pedhc.2011.03.011.

Quintas-Hijós, A., C. Peñarrubia-Lozano, and J. C. Bustamante. 2020. "Analysis of the Applicability and Utility of a Gamified Didactics with Exergames at Primary Schools: Qualitative Findings from a Natural Experiment." *PLoS ONE* 15 (4): e0231269. doi:10.1371/journal.pone.02311269.

Quintas, A. 2019. "Análisis del potencial didáctico de los exergames: reconceptualización y enfoque pedagógico." *Scholè. Rivista di educazione e studi culturali* 3 (1): 97–116.

Quintas, A. 2020. *Teoría educativa sobre tecnología, juego y recursos en didáctica de la educación infantil*. Zaragoza: Prensas de la Universidad de Zaragoza.

Quintas, A., J. C. Bustamante, F. Pradas, and C. Castellar. 2020. "Psychological Effects of Gamified Didactics with Exergames in Physical Education al Primary Schools: Results from a Natural Experiment." *Computers & Education* 152: 103874. doi:10.1016/j.compedu.2020.103874.

Rivoltella, P. C. 2017. *Media Education. Idea, metodo, ricerca*. Brescia, Italy: ELS La Scuola.

Robertson, J., R. Jepson, A. Macvean, and S. Gray. 2016. "Understanding the Importance of Context: A Qualitative Study of a Location-Based Exergame to Enhance School Childrens Physical Activity." *PLoS One* 11 (8): e0160927. doi:10.1371/journal.pone.0160927.

Ryan, R. M., and E. L. Deci. 2017. *Self-determination Theory: Basic Psychological Needs in Motivation, Development, and Wellness*. Nueva York: Guilford Press.

Sailer, M., J. U. Hense, S. K. Mayr, and H. Mandl. 2017. "How Gamification Motivates: An Experimental Study of the Effects of Specific Game Design Elements on Psychological Need Satisfaction." *Computers in Human Behavior* 69: 371–380. doi:10.1016/j.chb.2016.12.033.

Song, H., W. Peng, and K. M. Lee. 2011. "Promoting Exercise Self-Efficacy with an Exergame." *Journal of Health Communication* 16 (2): 148–162. doi:10.1080/10810730.2010.535107.

Staiano, A. E., and S. L. Calvert. 2011. "Exergames for Physical Education Courses: Physical, Social, and Cognitive Benefits." *Child Development Perspectives* 5 (2): 93–98.

Street, T. D., S. J. Lacey, and R. R. Langdon. 2017. "Gaming Your Way to Health: A Systematic Review of Exergaming Programs to Increase Health and Exercise Behaviors in Adults." *Games for Health Journal* 6 (3): 136–146. doi:10.1089/g4h.2016.0102.

Sun, H. 2013. "Impact of Exergames on Physical Activity and Motivation in Elementary School Students: A Follow-Up Study." *Journal of Sport and Health Science* 2 (3): 138–145. doi:10.1016/j.jshs.2013.02.003.

Torres, M. L. 2017. "Una mirada hacia un nuevo aprendizaje: el trabajo por proyectos." In *Temas clave de Educación Infantil (0-6 años)*, edited by O. Negrín and M. Barcia, 179–196. Madrid: Editorial Arco.

van Aart, I., E. Hartman, M. Elferink-Gemser, R. Mombarg, and C. Visscher. 2017. "Relations among Basic Psychological Needs, PE-Motivation and Fundamental Movement Skills in 9-12-Year-Old Boys and Girls in Physical Education." *Physical Education and Sport Pedagogy* 22 (1): 15–34. doi:10.1080/17408989.2015.1112776.

Verjans-Janssen, S. R. B., D. H. H. Van Kann, S. Gerards, S. B. Vos, M. W. J. Jansen, and S. P. J. Kremers. 2018. "Study Protocol of the Quasi-Experimental Evaluation of 'KEIGAAF': A Context-Based Physical Activity and Nutrition Intervention for Primary School Children." *BMC Public Health* 18: 842. doi:10.1186/s12889-018-5764-3.

Vernadakis, N., E. Zetou, V. Derri, E. Bebetsos, and F. Filippou. 2014. "The Differences Between Less Fit and Overweight Children on Enjoyment of Exergames, Other Physical Activity and Sedentary Behaviours." *Procedia - Social and Behavioral Sciences* 152: 802–807. doi:10.1016/j.sbspro.2014.09.324.

Watson, D., J. Adams, L. B. Azevedo, and C. Haighton. 2016. "Promoting Physical Activity with a School-Based Dance Mat Exergaming Intervention: Qualitative Findings from a Natural Experiment." *BMC Public Health* 16: 609. doi:10.1186/s12889-016-3308-2.

Williams, W. M., and C. G. Ayres. 2020. "Can Active Video Games Improve Physical Activity in Adolescents? A Review of RCT." *International Journal of Environmental Research and Public Health* 17 (2): 669. doi:10.3390/ijerph17020669.

Wu, X. Y., L. H. Han, J. H. Zhang, S. Luo, J. W. Hu, and K. Sun. 2017. "The Influence of Physical Activity, Sedentary Behavior on Health-Related Quality of Life among the General Population of Children and Adolescents: A Systematic Review." *PLoS One* 12 (11): e0187668. doi:10.1371/journal.pone.0187668.